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Prediction of chromatin loops in human genome using deep neural networks at the megabase scale

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Abstract. DNA sequence analysis has traditionally relied on methods like Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and hybrid models, but these often struggle to capture long-range dependencies. Recently, BERT-like architectures have shown promise by using self-attention mechanisms to detect complex patterns in DNA sequences. However, their ability to handle larger contexts is limited to kilobase scales, which can be a drawback for tasks needing broader genomic insights. This study explores HiCDiffusion, a model built to predict Hi-C contact maps at megabase resolution, and examines its potential to improve DNA sequence analysis by offering large-scale context. We specifically evaluate its performance in loop detection and compare its accuracy to leading methods, showing how it complements BERT-like models in capturing extensive genomic interactions.

Keywords: Artificial intelligence, Machine learning, BERT, Diffusion, Hi-C, Chromatin looping

1 Introduction

The spatial organization of chromatin is vital for gene regulation and genome functionality [10, 16]. Chromatin loops enable interactions between distant genomic regions, playing a crucial role in regulating gene expression [4]. However, predicting these loops is challenging due to the length of genomic sequences and incomplete nature of the data. Genomic data is derived from experimental assays such as Hi-C [17], ChIA-PET [8]. These methods introduce substantial noise and biases, leading to incomplete and sometimes contradictory chromatin interaction maps. The problem is compounded by the fact that most chromatin interaction datasets rely on probabilistic inferences rather than direct observations, making the labels inherently uncertain and variable across different cell types and conditions.

Many approaches combine DNA sequence features with external biological data to improve prediction accuracy [15, 24–26]. For instance, LoopPredictor

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[24] incorporates chromatin accessibility, transcription factor binding sites, and histone modifications, while deepTACT [15] integrates DNase-seq data to enhance sequence-based models. Although effective, these methods often rely on high-quality external annotations, adding complexity.

Convolutional Neural Networks (CNNs) [13] and Recurrent Neural Networks (RNNs) [6] are widely used for chromatin interaction prediction [1, 2, 15, 25]. Chromatin interaction neural network (ChINN) [1] utilizes CNNs for local pattern detection, and deepTACT [15] combines CNNs and RNNs to capture local and sequential dependencies. However, these architectures struggle with long genomic sequences and fail to capture broader contexts due to their localized or sequential processing. Transformer-based models, like DHL [3], inspired by DNABERT [12] address these limitations by leveraging self-attention for long-range dependencies but remain constrained by sequence length limitations.

HiCDiffusion [2] provides a new approach by generating high-resolution Hi-C interaction maps, offering large-scale genomic context. Initially developed for experimental Hi-C data replication, HiCDiffusion excels in tasks like Topologically Associating Domain (TAD) calling, capturing biologically significant features. Its high fidelity and minimal reliance on external data make it well-suited for chromatin loop prediction.

This study evaluates HiCDiffusion's predictive power and compares its performance with state-of-the-art methods, emphasizing its role as a robust tool for analyzing chromatin interactions and capturing long-range genomic dependencies.

2 Data

The dataset used in this study originates from the 4D Nucleome (4DN) project [21], featuring CTCF *in situ* ChIA-PET data from GM12878 cells. It includes alignment files (BAM) processed through the Hi-C Processing Pipeline, chromatin interaction data (BEDPE), and peak data (BED) from ChIA-PIPE results. BAM files specify the genomic locations of mapped reads, BEDPE files define chromatin interactions, and BED files highlight areas of significant CTCF enrichment. The reference genome used for alignment was hg38, sourced from the 1000 Genomes Project [18]. Together, these elements provide a solid foundation for identifying chromatin looping candidates and their genomic context.

An overview of the data processing workflow is illustrated in Figure 1. The following subsections describe in detail the dataset preprocessing and generation pipeline used in this study.

2.1 Data Generation Pipeline

Consensus DNA sequences were generated from BAM files using the `delly` variant caller [20] and `vcftools` [5], resulting in FASTA files. To identify stranded CTCF motifs (with forward or backward orientation), FIMO software [9] was applied to the hg38 reference genome and generated consensus sequences. Three

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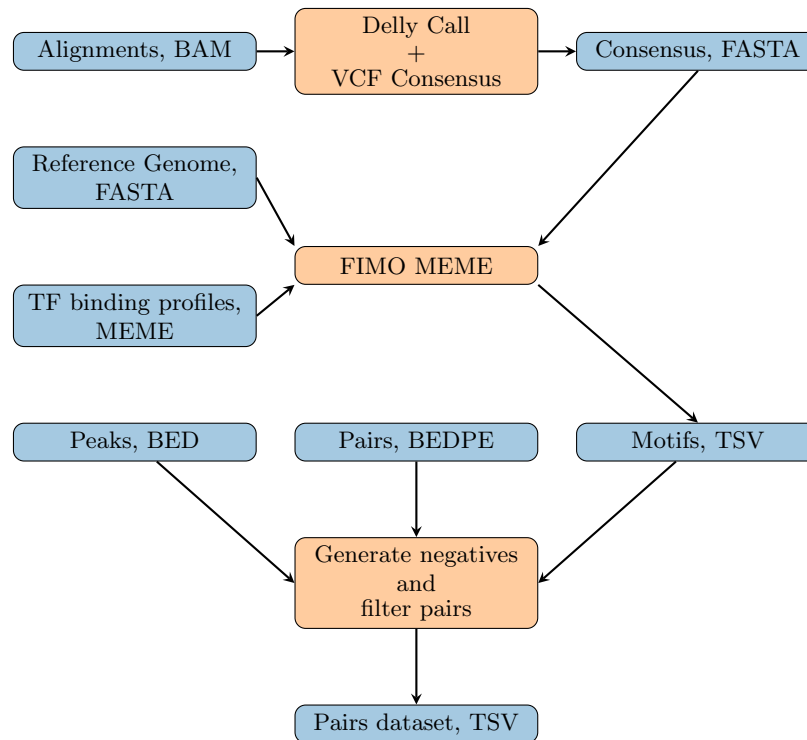


Fig. 1. Data preprocessing pipeline

motif versions³ from the JASPAR database [19] were used to ensure accurate detection of CTCF binding sites while accounting for orientation.

2.2 Data Integration and Labeling

CTCF motifs were integrated with ChIA-PET peak data to assign a dominant strand orientation to each peak. Peaks were labeled as forward or backward based on the majority motif orientation. Peaks with an equal number of forward and backward motifs were split into separate entries for each direction. BEDPE pairs were filtered by intersecting anchors with labeled peaks. Positive pairs were defined where left and right anchors overlapped forward and backward-directed peaks, respectively, with a PET (paired-end tag) count ≥ 3 . BEDPE pairs with PET counts < 3 were labeled as negatives, ensuring that positives reflected chromatin interactions supported by sufficient evidence [14].

Negative pairs were generated by randomly combining anchors from forward and backward-directed peaks, ensuring they matched the motif orientations of

³ MA0139.2.meme, MA1930.2.meme and MA1929.2.meme

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positive pairs. To reduce bias, negative pairs adjacent to positive ones were excluded. This ensured all pairs adhered to consistent labeling criteria and maintained biologically meaningful distinctions.

2.3 Final Filtering

Pairs were filtered to fit within the HiCDiffusion window (2,097,152 base pairs) and the 512-token input limit of DNABERT-S [27]. This preprocessing step ensured compatibility with both HiCDiffusion and DNABERT-S, aligning the dataset with their respective constraints for downstream analysis. To reduce confusion in the following sections, we refer to DNABERT-S simply as DNABERT. This naming convention distinguishes it from the original DNABERT model while maintaining clarity regarding the enhanced variant used in this study.

3 Model

The proposed model integrates sequence-based features with chromatin interaction context to predict chromatin loops. It combines outputs from DNABERT and HiCDiffusion to generate a comprehensive representation of anchor pairs. DNABERT encodes the left and right anchor sequences separately. Each sequence is processed through DNABERT, and the resulting embeddings are mean pooled, producing two 768-dimensional vectors that capture the local genomic features of the left and right anchors.

HiCDiffusion provides extended genomic context by processing a one-hot encoded sequence that spans both anchors. It generates a high-resolution Hi-C interaction matrix (256×256), which is combined with a boolean mask. The mask highlights the positions of the left and right anchors on the matrix using non-zero values in specific rows and columns (visualized as orange and blue lines in the model diagram, see Figure 2). The resulting $256 \times 256 \times 2$ matrix, with separate channels for the Hi-C data and mask, is processed through 2D residual convolutional blocks, extracting spatial interaction features. This produces a 768-dimensional vector representing the broader genomic context of the anchor pair.

The three 768-dimensional vectors – two from DNABERT (for both anchors) and one from HiCDiffusion – are concatenated to form a unified representation of the anchor pair. This combined vector is passed through fully connected layers to classify whether the anchor pair forms a chromatin loop.

3.1 Training

Several data augmentation strategies were used to enhance model robustness. The context window, containing both anchors, was randomly shifted within the global sequence to vary anchor positions (e.g., at the beginning, center, or end). Context windows and anchor subsequences were randomly sourced from either the reference genome or consensus sequences to introduce variability. Positive

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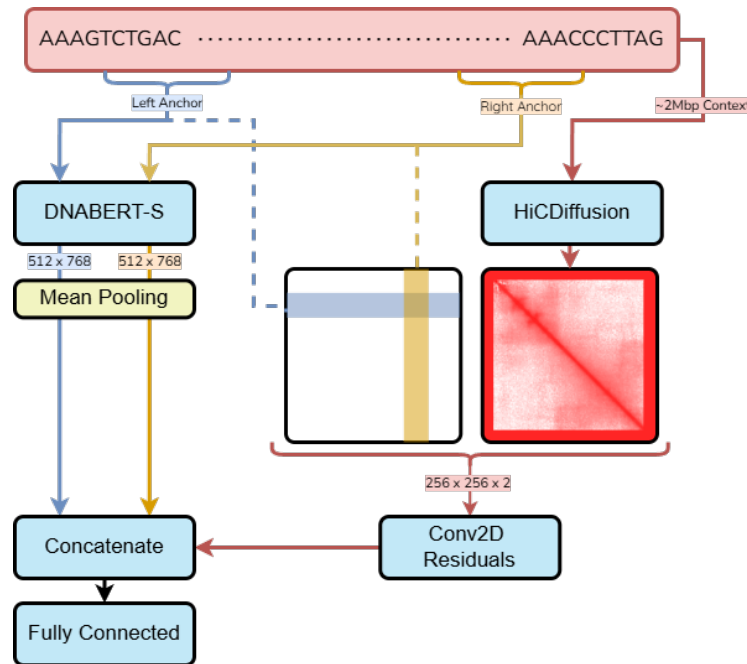


Fig. 2. Model architecture diagram

and negative samples were drawn equally to maintain class balance. During evaluation, random shifts of context windows were disabled, and anchor pairs were always centered within the sequence to ensure consistent evaluation conditions.

The model was trained using the AdamW optimizer with a 10% dropout rate. The learning rate was scheduled with a warmup over the first 5% of steps, starting at 0 and increasing to 5e-5, followed by a linear decay to zero. Training used a batch size of 8, with binary cross-entropy as the loss function. The best model was selected based on the average precision score. F1 score, recall, precision, and ROC-AUC were also tracked during training to comprehensively evaluate performance.

For consistency with prior work, we used the same train-test splitting strategy as the original HiCDiffusion development, validating the pretrained model on chromosome 15 and testing it on chromosome 14. This ensured fair evaluation while leveraging the established capabilities of HiCDiffusion.

To improve generalization and mitigate overfitting, dropout layers were included in the feedforward classifier. A dropout rate of 10% was applied to the dense layers, randomly zeroing a fraction of input units during training [23]. This regularization strategy promotes robustness by preventing co-adaptation of neurons and encourages the network to learn more generalizable patterns.

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3.2 ChINN

For our study, we selected the Chromatin Interaction Neural Network (ChINN) model as it most closely aligns with our experimental setup. ChINN operates exclusively on DNA sequence input and is designed to classify CTCF-mediated chromatin loops, making it particularly suited to our research objectives.

The ChINN model was trained using a combination of datasets. Specifically, we used BEDPE interaction pairs and BED peaks derived from our dataset (4DN) and supplemented them with DNase hypersensitivity data from the ENCODE project [7] (ENCFF759OLD [22]).

We also attempted to utilize the original dataset. However, it was aligned to the hg19 reference genome. To ensure consistency with our model, which was trained on the hg38 reference, we uplifted all BED and BEDPE data using the UCSC Genome Browser's LiftOver tool⁴. Unfortunately, this resulted in significantly degraded performance, likely due to noise introduced during the coordinate conversion process.

To maintain consistency with our overall train-test split, both versions of the trained model were validated on chromosome 15 and tested on chromosome 14⁵.

Figure 3 provides a qualitative comparison of chromatin loop predictions from multiple models, facilitating visual assessment of their alignment with Hi-C contact frequencies. In the following section, we quantify these observations using evaluation metrics and benchmark performance.

4 Results

To evaluate the proposed model, we conducted experiments using real Hi-C matrices from the 4D Nucleome (4DN) project and predicted Hi-C matrices generated by HiCDiffusion. This comparison assessed the impact of using HiCDiffusion as a contextual module against experimentally derived Hi-C data. The test set, consisting of data from chromosome 14, included 6,680 candidate pairs, of which 214 (approximately 3.2%) were positives. For consistency, sequences were derived from the reference genome, and the context window was centered.

The results on Table 1 highlight the value of contextual information. DNABERT alone showed the least performance across the metrics. Adding real Hi-C matrices improved recall and F1, creating a more robust model. Similarly, HiCDiffusion-enhanced matrices boosted recall and F1 compared to DNABERT alone, achieving results close to those with real Hi-C data. The slightly lower performance of HiCDiffusion-based predictions reflects the limitations of generative models in fully replicating experimental data. However, its comparable results demonstrate HiCDiffusion's utility, especially when real Hi-C data is unavailable. This shows the potential of HiCDiffusion to provide effective contextual information for chromatin loop prediction.

⁴ <https://genome.ucsc.edu/cgi-bin/hgLiftOver>

⁵ All modifications and scripts used in this process are based on a forked version of the ChINN repository: <https://github.com/SFGLab/ChINN>.

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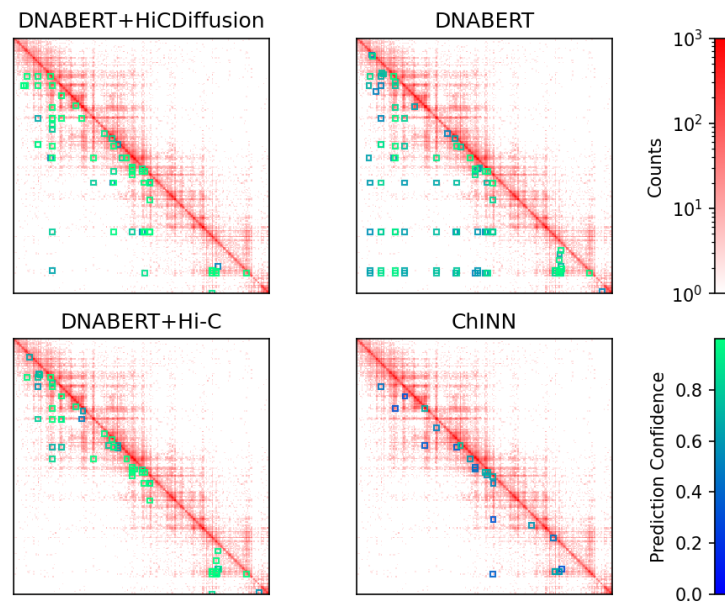


Fig. 3. Comparison of predicted chromatin loops from different models overlaid on the ground truth Hi-C contact map for the chr14:22Mbp–24Mbp region. The red heatmap represents Hi-C log-scaled interaction counts, while squares indicate predicted loops, with their color corresponding to the confidence score.

4.1 Predicting chromatin interactions from open chromatin regions

To assess the predictive performance of our proposed models in comparison with the ChINN baseline, we conducted a benchmark on candidate chromatin loop interactions derived from open chromatin regions. The ChINN model was originally trained and evaluated on hypersensitive sites, representing accessible regions of the genome, making this evaluation a natural extension of its original use case. To ensure comparability, we used DNase-seq data from the same data source used in the training of ChINN – to define open chromatin regions. These regions were extended by an additional 1000 base pairs to allow for potential interactions and overlapping anchor identification, resulting in a set of 1014 candidate region pairs for chromosome 14. Within this set, approximately 4.7% of the pairs were labeled as positives based on experimentally supported interactions. This evaluation provides a consistent and relevant setting to test the performance of our models alongside ChINN on the biologically meaningful subset of accessible chromatin.

Figure 5 shows the distribution of predicted confidence scores stratified by anchor distance. DNABERT alone maintains high confidence across all distances, but with narrow interquartile ranges, indicating over-reliance on sequence features and limited sensitivity to increasing genomic span. DNABERT+Hi-C

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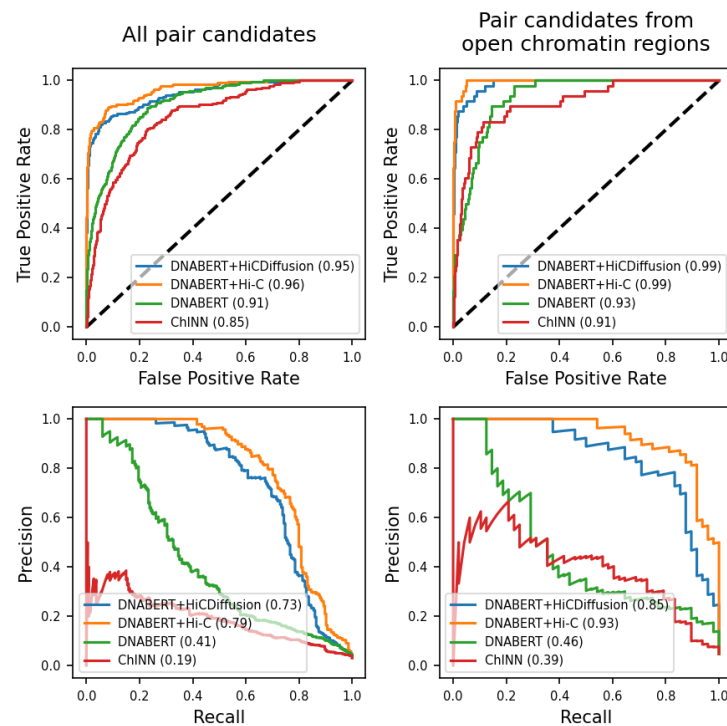


Fig. 4. ROC and Precision-Recall curves for all test pair candidates (left column) and the subset restricted to open chromatin regions (right column).

moderates this effect, producing more distance-sensitive predictions, particularly in the 0.5–1.5 Mbp range, where Hi-C data enriches spatial resolution. DNABERT+HiCDiffusion, while displaying broader variability and reduced overall confidence, maintains meaningful differentiation across distance bins, especially at long ranges, highlighting its capacity to model distal interactions in the absence of experimental Hi-C input. The ChINN model exhibits relatively flat confidence distributions across all distance bins.

In Figure 6, prediction confidence is plotted against PET support, a proxy for interaction strength. DNABERT-based models demonstrate strong monotonic trends with increasing PET counts, with DNABERT+Hi-C achieving near-saturation of confidence beyond 5 PETs. DNABERT+HiCDiffusion follows a similar trajectory, with slightly reduced confidence at low support levels. In contrast, both ChINN variants exhibit lower and more variable confidence trends across PET counts, reflecting weaker calibration to quantitative interaction strength. These results support the improved sensitivity and interpretability of DNABERT-based models on open chromatin-derived candidates, particularly when augmented with Hi-C or generative spatial context.

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Table 1. Performance comparison of models for chromatin loop prediction on chromosome 14.

Configuration	Precision	Recall	F1	ROC-AUC	Average Precision
ChINN	0.220	0.388	0.281	0.853	0.193
DNABERT	0.421	0.360	0.388	0.906	0.410
DNABERT+Hi-C	0.655	0.771	0.708	0.964	0.785
DNABERT+HiCDiffusion	0.620	0.738	0.674	0.947	0.735

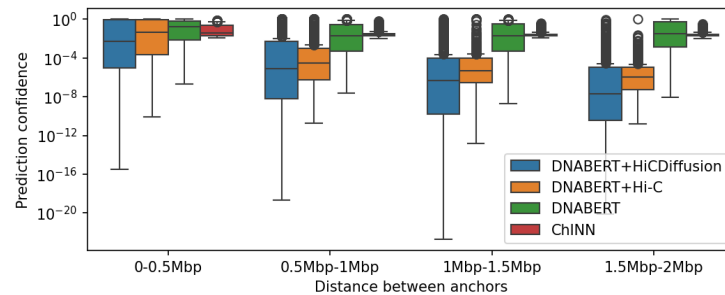


Fig. 5. Prediction confidence by genomic distance bins and configurations. The figure illustrates differences in prediction confidence across configurations.

The bar plot on Figure 7 represents the values of the confusion matrix, providing insight into the classification performance of different chromatin loop prediction models. The models are grouped into three categories: DNABERT, DNABERT+Hi-C, DNABERT+HiCDiffusion and ChINN.

Across all models, the true negative count remains consistently high, suggesting that each model is effective at filtering out non-looping regions with high confidence—even in the absence of explicit Hi-C input.

True positive rates are comparable for DNABERT+HiCDiffusion and DNABERT+Hi-C, both outperforming the base DNABERT and ChINN mod-

Table 2. Performance comparison of models for chromatin loop prediction on chromosome 14 from open regions.

Configuration	Precision	Recall	F1	ROC-AUC	Average Precision
ChINN	0.443	0.562	0.495	0.907	0.404
DNABERT	0.338	0.458	0.389	0.930	0.461
DNABERT+Hi-C	0.840	0.875	0.857	0.995	0.927
DNABERT+HiCDiffusion	0.677	0.875	0.764	0.986	0.850

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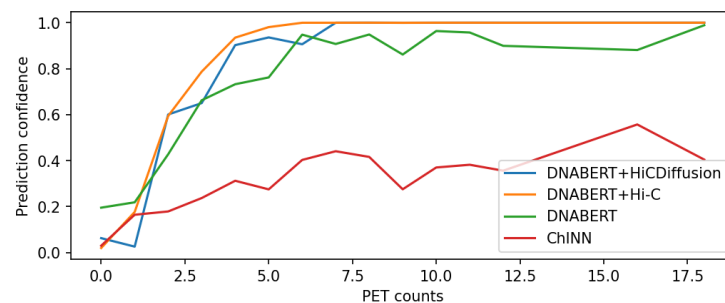


Fig. 6. Average prediction confidence by PET count

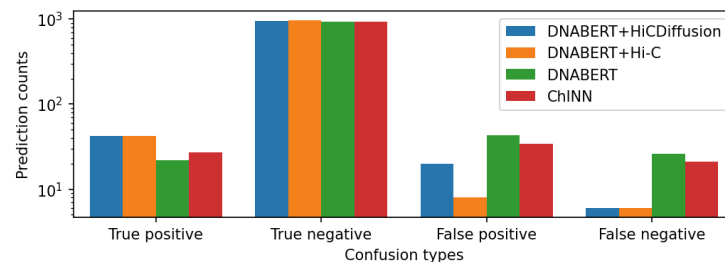


Fig. 7. Confusion types on chromosome 14

els. This highlights the benefit of incorporating Hi-C data, whether directly or through generative modeling, in enhancing the detection of true chromatin loops.

The most substantial differences between models appear in the false positive and false negative counts. DNABERT+Hi-C has the lowest number of incorrect predictions overall. DNABERT+HiCDiffusion shows a higher false positive count, which may be due to some noise in the generated HiCDiffusion predictions leading to overly confident predictions. In contrast, both DNABERT and ChINN have higher rates of false positives and false negatives, indicating lower precision and recall. These results highlight the benefit of incorporating Hi-C information to improve prediction accuracy and reduce errors in identifying both real and non-existent interactions.

These results demonstrate that while baseline DNABERT can differentiate chromatin interactions to some extent, incorporating Hi-C data significantly enhances its predictive accuracy. Correspondingly, the performance metrics summarized in Table 2 show consistent improvements in precision, recall, and F1 score. Furthermore, the area under the ROC and PR curves confirms the benefit of hybrid models (see Figure 4), with higher AUC values reflecting stronger discriminative capability. Altogether, these findings suggest that embedding Hi-C-derived constraints enables more robust and reliable identification of functional chromatin loops.

4.2 3D Convolutional Encoder

In addition to the 2D residual convolutional encoder used for Hi-C image processing, we implemented an experimental variant using a 3D convolutional encoder. The goal was to enable the model to jointly process both the Hi-C image and its corresponding binary mask as a volumetric input. The mask indicates the anchor pair positioning relative to Hi-C image. This helps guide the encoder's attention to the relevant interaction region and suppresses misleading signals from nearby TAD structures or off-target contacts.

To support this, we reformulated the original two-channel input of shape $256 \times 256 \times 2$ into a pseudo-3D tensor with shape $256 \times 256 \times 1 \times 2$, treating the Hi-C contact map and the mask as two adjacent slices along a synthetic depth dimension. This configuration allows 3D convolutional kernels to jointly learn spatial features and inter-channel relationships, potentially improving early fusion of structural and task-specific cues.

We evaluated this 3D encoder exclusively on the pretraining classification task, without downstream modules such as DNABERT. Performance was benchmarked against the baseline 2D encoder. Notably, average precision improved from 0.03 (2D encoder with predicted Hi-C) to 0.08 using the 3D encoder. When real Hi-C images were used, performance further increased to 0.20. These results suggest that 3D convolutional modeling can enhance the network's ability to isolate relevant contact regions and learn more effective representations for chromatin interaction classification.

5 Discussion

Our results demonstrate that integrating HiCDiffusion with DNABERT significantly enhances performance in chromatin loop prediction, particularly in recall and F1 score, compared to sequence-only approaches. DNABERT alone shows high precision, especially for short-range interactions, but its reliance solely on anchor sequences limits its ability to generalize across broader genomic contexts and reduces recall at longer distances. Incorporating real Hi-C data provides balanced confidence calibration and improved performance across all metrics. HiCDiffusion, while showing slightly greater variability in prediction confidence than real Hi-C, still offers consistent improvements over the sequence-only baseline and proves to be a viable alternative when experimental data is not available.

To benchmark against existing models trained specifically on open chromatin, we evaluated all configurations on a subset of candidate pairs derived from DNase-defined accessible regions, aligning with the original training setup of ChINN. On this subset, our models demonstrated significantly stronger performance in terms of prediction confidence and distance-aware behavior, outperforming ChINN. Furthermore, this strong performance was consistent across both the full dataset of candidate interactions and the open chromatin-derived subset, reinforcing the generalizability and robustness of our approach.

There is still room for improvement. Our experiments were conducted using data from a single cell line, which limits broader applicability. Incorporating data

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from multiple cell types could help capture cell-specific chromatin architecture. Moreover, decoupling the DNABERT encoders for left and right anchors could allow better modeling of asymmetric sequence features. Encoding the sequence between anchors using a sliding window [11] might also provide additional contextual information, improving detection of distal interactions.

6 Conclusion

This study presents a robust chromatin loop prediction framework that combines DNABERT with HiCDiffusion-generated contact maps. The approach achieves performance comparable to models using real Hi-C data, and consistently outperforms the ChINN baseline on both the full set of candidate pairs and the subset derived from open chromatin regions. This highlights the advantage of combining transformer-based sequence modeling with large-scale spatial context, whether derived experimentally or via generative modeling.

Our analysis of prediction confidence across genomic distance and PET support further confirms the effectiveness of fusion of DNABERT with Hi-C data or HiCDiffusion predictions in producing interpretable and distance-aware predictions. The model's ability to generalize across full genome-wide candidate sets and open region subsets underscores its flexibility and potential for broader application.

In parallel, we explored a 3D convolutional encoder capable of jointly processing Hi-C and mask inputs as volumetric data. Preliminary results on the encoder pretraining task suggest promising improvements in representation learning, pointing toward a potential upgrade for future versions of the full model.

All code, trained models, and evaluation data are available at <https://github.com/SFGLab/HiCDiffusionLooping>. Future work may explore extending this framework across multiple cell types, refining asymmetric anchor encoding, incorporating inter-anchor sequence features, and integrating the 3D encoder for enhanced joint modeling of genomic structure and context.

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